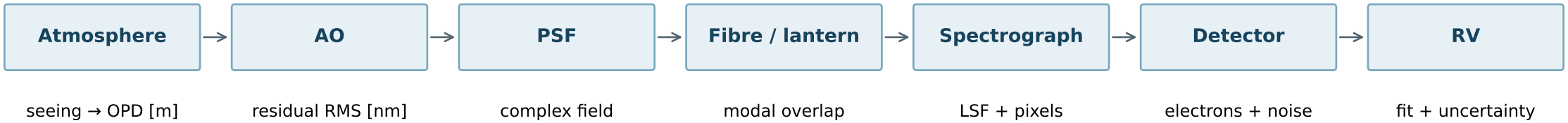A reproducible chain from wavefront error to RV information